Lecture 12 - Inverse and Implicit Differentiation

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10/20/2026

Updates

Curve information forthcoming - in contact with course coordinator about adjusting final grade distribution.
Regrade requests for Exam 1 closed.

Quiz Time!

Let's get into neat rows, spread out, and slay this quiz!

Inverse and Implicit Differentiation

The Road So Far

Recall we're trying to build a *synthetic* approach to studying derivatives:
Atomic Rules:

- Power Rule: $\frac{d}{dx}x^n$
- Trigonometric Rules: $\frac{d}{dx}\sin(x)$ etc.
- Exponential/Logarithm Rules: $\frac{d}{dx}e^x$, $\frac{d}{dx}\ln(x)$

Combination Rules:

- Sum/Difference Rule: $\frac{d}{dx}(f(x) \pm g(x))$
- Product Rule: $\frac{d}{dx}f(x)g(x)$
- Chain Rule: $\frac{d}{dx}g(f(x))$
- Constant Multiple Rule: $\frac{d}{dx}c \cdot f(x)$
- Quotient Rule: $\frac{d}{dx}\frac{f(x)}{g(x)}$
- Inverse Function Rule: $\frac{d}{dx}f^{-1}(x)$

Explicit vs. Implicit Definitions I

Up until now we've been dealing with *explicit* definitions of various functions and have been using those *explicit* definitions to compute their derivatives.
An *explicit* definition of a function is one where we defined $f(x)$ by *explicitly* saying where a given input x maps to. E.g.:

- $f(x) = x^2$
- $g(x) = \sin(x)$
- $h(x) = \sqrt{x^2 + 2x}$
- $j(x) = e^{\sin(x)}$
- $k(x) = (x + \sin(x))^{\frac{1}{2}}$
- etc

This is not the only way in which we can define functions!

Explicit vs Implicit Definitions II

We could instead define relations *implicitly* - i.e. by writing out some equation that we want the function to satisfy, e.g.:

- $x^2 + y^2 = 1$
- $y^2 = x^3 + 2x + 1$
- $\arctan(\frac{y}{x}) = \frac{\pi}{4}$
- $\sin(x) + \cos(y) = 1$
- $x^2 - y^2 = 2$
- $x^3 + y^3 = 1$

Just as we can *graph* a function: i.e. we plot all the points $(x, f(x))$, we can plot some implicit equation $F(x, y) = 0$ by plotting all the points (x, y) so that $F(x, y) = 0$.
When is this useful?

Applications of Implicit Relations

Oftentimes, real world process don't have simple ways of relating one quantity to another quantity.
Consider: conservation of momentum/energy in physics:

$$\sum_{i=1}^{n} m_i v_i^{init} = \sum_{i=1}^{n} m_i v_i^{final} \quad (1)$$

$$\sum_{i=1}^{n} \frac{1}{2} m_i (v_i^{init})^2 = \sum_{i=1}^{n} \frac{1}{2} m_i (v_i^{final})^2 \quad (2)$$

The relation between initial and final velocities here are not a simple input-to-output relationship - they are equations that need to be enforced.
Also: predator-prey models, resource allocation, budgeting, etc.

Inverse Functions as a Relation

To address our final desired synthetic rule (inverse function rule) we can also see that inverse functions are defined via an implicit rule:

Inverse Function

For a function $f(x)$ with domain $D \subseteq \mathbb{R}$ and range $R \subseteq \mathbb{R}$, the inverse function $f^{-1}(x)$ is the function that:

- Has domain R (domain of inverse is range of original)
- Has range D (range of inverse is domain of original)
- Satisfies the two relations:
  1. $f(f^{-1}(x)) = x$
  2. $f^{-1}(f(x)) = x$

  i.e. $f(x)$ and $f^{-1}(x)$ "undo" eachother.

Examples of Inverse Functions

For some examples:

1. If $f(x) = x^2$ with domain *the positive real numbers* and range *the positive real numbers* $f^{-1}(x) = \sqrt{x}$ is the positive square root.
2. If $f(x) = x^2$ with domain *the negative real numbers* and range *the positive real numbers* $f^{-1}(x) = -\sqrt{x}$ is the negative square root.
3. If $f(x) = \sin(x)$ with domain $[0, 2\pi)$ and range $[-1, 1]$ $f^{-1}(x) = \arcsin(x)$
4. If $f(x) = e^x$ with domain $\mathbb{R}$ and range $(0, \infty)$ $f^{-1}(x) = \ln(x)$.

Inverse Differentiation I

Question: Using the relation

$$f(f^{-1}(x)) = x \tag{3}$$

how might we relate the derivative of $f(x)$ with the derivative of it's inverse function $f^{-1}(x)$?
Answer: Chain Rule!

$$f(f^{-1}(x)) = x \tag{4}$$

Differentiate both sides (5)

$$\frac{d}{dx} f(f^{-1}(x)) = \frac{d}{dx} x \tag{6}$$

$$\left(\frac{d}{dx} f^{-1}(x)\right) \cdot f'(f^{-1}(x)) = 1 \tag{7}$$

$$\frac{d}{dx} f^{-1}(x) = \frac{1}{f'(f^{-1}(x))} \tag{8}$$

Inverse Differentiation II

Inverse Differentiation

If $f(x)$ and $f^{-1}(x)$ are an inverse pair of functions then:

$$\frac{d}{dx}f^{-1}(x) = \frac{1}{f'(f^{-1}(x))} \tag{9}$$

Inverse Differentiation III

Ex. $\frac{d}{dx}\ln(x)$.
$\ln(x)$ is inverse to $e^x \implies f(x) = e^x, f^{-1}(x) = \ln(x)$.

$$\frac{d}{dx}\ln(x) = \frac{1}{\left(\frac{d}{du}e^u\right)|_{u=\ln(x)}} = \frac{1}{(e^u)|_{u=\ln(x)}} \tag{10}$$

$$= \frac{1}{e^{\ln(x)}} = \frac{1}{x} \tag{11}$$

Derivative of Natural Logarithm

$$\frac{d}{dx}\ln(x) = \frac{1}{x} \tag{12}$$

Inverse Differentiation IV

Ex. $\frac{d}{dx}\arcsin(x)$.

$$\frac{d}{dx}\arcsin(x) = \frac{1}{\left(\frac{d}{du}\sin(u)\right)|_{u=\arcsin(x)}} = \frac{1}{(\cos(u))|_{u=\arcsin(x)}} \tag{13}$$

$$= \cos(\arcsin(x)) = \frac{1}{\sqrt{1-x^2}} \tag{14}$$

$$\frac{d}{dx}\arccos(x) = \frac{1}{-\sin(\arccos(x))} = -\frac{1}{\sqrt{1-x^2}} \tag{15}$$

$$\frac{d}{dx}\arctan(x) = \frac{1}{\sec^2(\arctan(x))} = \frac{1}{1+x^2} \tag{16}$$

Inverse Differentiation V

Derivatives of Inverse Trigonometric Functions

$$\frac{d}{dx}\arcsin(x) = \frac{1}{\sqrt{1-x^2}} \tag{17}$$

$$\frac{d}{dx}\arccos(x) = -\frac{1}{\sqrt{1-x^2}} \tag{18}$$

$$\frac{d}{dx}\arctan(x) = \frac{1}{1+x^2} \tag{19}$$

Implicit Differentiation I

What about for the other kinds of implicit relations?

Q: At a point (x, y) on the circle of radius 1, what is the slope of the line tangent to the circle at that point?

Circle of radius 1 $\implies x^2 + y^2 = 1$.

$$x^2 + y^2 = 1 \tag{20}$$

Differentiate both sides w.r.t. x (21)

$$\frac{d}{dx}(x^2 + y^2) = \frac{d}{dx}1 \tag{22}$$

$$\frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) = 0 \tag{23}$$

$$2x + \frac{d}{dx}(y(x)^2) = 0 \tag{24}$$

$$2x + 2y \cdot \frac{dy}{dx} = 0 \quad 2y\frac{dy}{dx} = -2x \quad \frac{dy}{dx} = -\frac{x}{y} \tag{25}$$

Implicit Differentiation II

The core idea is that in an implicit relation $F(x, y) = 0$, we can *always* treat one of the variables as a function of the other (fancy reason: *implicit function theorem*) and thus can differentiate implicit relations just by using the chain rule!
We'll see more application oriented examples when we do related rates soon :)

Worksheet Time

Let’s get into groups and work on the worksheet!

End of Class

That’s all folks!